MECHALINE DATA ANALYTICS INTERNSHIP PROGRAM

WEEK 1 ASSIGNMENT

Project: Pakistan Used Car Market Intelligence & Advanced Data Analysis

Assigned To: Muneeb

Duration: 5 Working Days

Project Objective

The objective of this project is to analyze Pakistan's used car market using the PakWheels dataset to identify pricing patterns, market trends, consumer preferences, and the factors influencing used vehicle prices. The intern will apply data cleaning, exploratory data analysis (EDA), descriptive analytics, diagnostic analytics, statistical analysis, and business intelligence techniques to generate actionable insights.

Day 1 – Data Understanding & Preparation

Tasks

- Study the complete dataset and understand every feature.
- Prepare a Data Dictionary describing each column.
- Identify missing values and inconsistent records.
- Remove duplicate entries.
- Handle null values using appropriate techniques.
- Correct inconsistent categorical values.
- Convert columns into appropriate data types.
- Identify and remove obvious outliers where justified.

Deliverables

- Cleaned Dataset
- Data Dictionary
- Data Cleaning Report

Day 2 – Exploratory Data Analysis (EDA)

Perform descriptive analysis on:

- Vehicle Brands
- Vehicle Models
- Manufacturing Year
- Fuel Type
- Transmission Type
- Engine Capacity
- Mileage
- Vehicle Price
- City-wise Listings

Create appropriate visualizations including:

- Histograms
- Bar Charts
- Box Plots
- Pie Charts
- Scatter Plots

Summarize the key observations from each visualization.

Day 3 – Advanced Analytics

Perform the following analyses:

Correlation Analysis

Determine relationships between:

- Price vs Mileage
- Price vs Manufacturing Year
- Price vs Engine Capacity

Present findings using a correlation matrix.

Segmentation Analysis

Segment vehicles into categories based on:

- Budget

- Mid-range
- Premium
- Luxury

Analyze:

- Brand distribution within each segment
- Popular models within each segment
- Average mileage
- Average manufacturing year

Market Trend Analysis

Identify:

- Top 10 Brands
- Top 10 Models
- Fastest-growing vehicle categories
- Fuel preferences
- Transmission preferences

Geographic Analysis

Analyze:

- Cities with highest listings
- Average vehicle prices by city
- Brand popularity by city

Day 4 – Business Intelligence

Answer the following business questions:

- Which brands dominate Pakistan's used car market?
- Which brands retain higher resale value?
- Which cities have the largest used car markets?
- Which vehicle features most influence selling price?
- Which fuel type is becoming more popular?
- Which transmission type dominates the market?
- What customer preferences can Mechaline leverage in the future?

Provide business recommendations supported by data.

Day 5 – Dashboard Development

Develop an interactive Power BI Dashboard containing:

KPI Cards

- Total Listings
- Average Price
- Median Price
- Average Mileage
- Most Popular Brand
- Most Popular Model
- Most Common Fuel Type
- Most Common Transmission

Dashboard Visualizations

- Brand Distribution
- Model Distribution
- Price Distribution
- Mileage Distribution
- Fuel Type Analysis
- Transmission Analysis
- City-wise Listings
- Manufacturing Year Trend
- Correlation Heatmap
- Price vs Mileage Scatter Plot
- Price by Brand
- Vehicle Segmentation Analysis

Final Deliverables

- Cleaned Dataset
- Data Dictionary
- Exploratory Data Analysis Report
- Advanced Analytics Report
- Business Recommendations Report
- Interactive Power BI Dashboard (.pbix)
- Final Presentation (8–10 Minutes)

Learning Outcomes

Upon successful completion of this project, the intern should be able to:

- Perform real-world data cleaning and preprocessing.
- Conduct exploratory and diagnostic data analysis.
- Apply correlation and segmentation techniques.
- Build business intelligence dashboards using Power BI.
- Extract actionable business insights from automotive market data.
- Present findings professionally to technical and non-technical stakeholders.